

Section 34: Network Device Management

Layer: Multi | Reference: Odom Vol2 Ch29-30: Network Management Protocols and Practices | *Original study notes — CCNA 200-301*

Beyond initial configuration, ongoing network operations rely on standardized practices for software lifecycle management, configuration backup, and license management - operational topics the CCNA exam expects you to recognize.

Key Concepts

- IOS image management: routers/switches store the running IOS image in Flash memory; upgrading typically means copying a new image to Flash and changing the boot variable to point to it
- 'boot system' commands define the order in which the device tries to load IOS images, providing a fallback if the primary image is corrupted
- Configuration backup should be automated and stored off-device (TFTP/SFTP/SCP server) so a hardware failure doesn't also destroy the only copy of the config
- TFTP (Trivial File Transfer Protocol) is simple and widely supported for IOS/config transfers but transmits without encryption - acceptable on a trusted management network, not over the open internet
- SCP/SFTP provide encrypted file transfer alternatives to TFTP when security requirements are higher
- Licensing models on modern Cisco platforms (e.g., smart licensing) require periodic check-in with Cisco's licensing servers (directly or via satellite/on-prem server) to remain compliant
- Configuration management tools (beyond raw CLI) like Cisco DNA Center, Ansible, or simple scripted backups via SSH increasingly automate what used to be manual per-device CLI work
- Network Time Protocol (NTP), syslog, and SNMP (covered in Section 15) are the day-to-day monitoring backbone that device management relies on for visibility
- Out-of-band (OOB) management (a dedicated management network/console server) ensures admins can still reach a device even if the production network itself is down

Command Reference

Command	Purpose
show flash:	List IOS images and other files stored in Flash memory
show version	Display the currently running IOS version, uptime, and hardware details
boot system flash:<filename>	Set the IOS image to load on next boot
copy tftp: flash:	Download a new IOS image from a TFTP server into Flash
copy running-config tftp:	Back up the running configuration to a TFTP server
archive config	Trigger an automatic configuration archive if the archive feature is configured

Command	Purpose
<code>show license / show license status</code>	View current licensing state on platforms using smart licensing

Configuration Walkthrough

Upgrade IOS: copy new image to Flash, set boot variable, reload

```
R1#copy tftp: flash:
Address or name of remote host []? 10.10.10.10
Source filename []? c2900-universalk9-mz.SPA.bin
R1(config)#boot system flash:c2900-universalk9-mz.SPA.bin
R1#copy running-config startup-config
R1#reload
```

Automate periodic configuration backups

```
R1(config)#archive
R1(config-archive)#path tftp://10.10.10.50/configs/r1-config
R1(config-archive)#write-memory
```

Worked Scenario

Q: A router's primary IOS image becomes corrupted after a failed upgrade, and the device fails to boot normally. What design choice would have prevented a full outage?

Keeping a known-working IOS image in Flash and configuring multiple 'boot system' statements (pointing to the new image first, with the old verified-working image as a fallback) lets the router automatically fall back to the working image if the primary fails to load - avoiding a full outage requiring physical console recovery via ROMMON.

Common Mistakes / Gotchas

- Upgrading IOS without keeping a fallback image and boot statement in place
- Relying solely on running-config without realizing configuration backups need to happen on a schedule, off-device, not just 'when I remember'
- Treating TFTP as fully secure when it has no encryption - acceptable for internal trusted segments, not as a general practice

Exam Tips

- ★ Know the practical difference between TFTP (simple, unencrypted) and SCP/SFTP (encrypted) for file transfer
- ★ Understand why off-device, automated config backups matter operationally, even though it's a 'soft' topic compared to protocol configuration